BAMA580A: Experiments & Causal Inference
Section: BA2

Group Project Report

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Instructor: Yann Cornil, David Hardisty
TAs: Rob Velzeboer, Shangwen Yi
Group 18:
Candice Miao 12978789
Romee Sun 99320285
Xiaoxi Xu 12491015
Michael Ruizhe Zhang 82889767
Sisly Lyu 72710643

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1. Executive summary

This project explores how message framing, either emphasizing durability or sustainability, affects consumer responses to a Stanley Cup product. We conducted a between-subjects survey experiment with 302 participants, who were randomly assigned to one of the two message conditions. Participants then rated the product across three dimensions: perceived performance, purchase intention, and likelihood to recommend. We also measured each participant's environmental concern level to assess potential moderation effects. Our analysis shows that sustainability framing significantly increased perceived product performance, although it did not significantly affect purchase intention or recommendation overall. Importantly, environmental concern significantly moderated the effect of framing on purchase intention, with the sustainability message being more persuasive among participants with stronger environmental values. A three-way interaction between framing, gender, and age was also significant, suggesting that framing effects differ across demographic segments. These findings imply that tailoring product messages to match consumer values and characteristics can enhance marketing effectiveness, especially for products with both functional and ethical appeal.

2. Background & Hypotheses: Framing Effect

2.1 Background

Framing is a powerful communication tool in marketing, where the same product benefit can be presented in different ways to influence consumer perception. In recent years, brands have increasingly used sustainability framing to appeal to eco-conscious consumers. However, for products that also have strong functional attributes, like the Stanley Cup, there is a trade-off between emphasizing ethical appeal (sustainability) and practical value (durability). Our study is inspired by past research on message framing, particularly in pro-social behavior and product evaluations, and aims to explore which message is more effective depending on consumer characteristics.

2.2 Hypotheses

- H1 (Main Effect): Sustainability framing will increase perceived product quality, purchase intention, and recommendation likelihood compared to durability framing. That is, participants who see the sustainability message are expected to rate the product more positively and show stronger behavioral intentions.
- H2 (Moderation Effect): The positive effect of sustainability framing on purchase intention will be stronger among participants with higher levels of environmental concern. In other words, people who already care about sustainability are expected to be more influenced by green messaging.
- H3 (Demographic Interaction): The framing effect will vary across demographic groups, especially age and gender. We hypothesize that younger consumers and women may be particularly responsive to sustainability framing, reflecting broader trends in eco-conscious purchasing behavior.

These hypotheses aim to test not just whether framing matters but for whom it matters most, providing useful insights for targeted marketing strategies.

3. Methodology

To examine the effects of message framing on consumer perception and behavior, we designed a between-subjects experiment using Qualtrics. Participants were randomly assigned to view one of two product slogans for the same Stanley Cup tumbler, representing two different message frames: a sustainability frame ("Built to refill, not to replace") and a durability frame ("Built to last, not to crack"). Each condition had approximately 151 participants, ensuring balanced group sizes for comparison.

Following the slogan exposure, participants responded to a series of outcome measures designed to capture their attitudes and intentions. We measured three key dependent variables:

- Perceived Performance, assessed through agreement with the statement:
  "*I believe this product would perform well in daily use.*"
- Purchase Intention, measured by the response to:
  "*I am willing to purchase this Stanley tumbler*."
- Recommendation Likelihood, captured by
  "*I will recommend this product to a friend or family member*."

To explore potential moderation effects, we also collected data on three key moderators:

- Gender Identity, with response options of Woman, Man, and Other.
- Generation (Age Group) categorized into Gen Z (18-26), Millennial (27-42), Gen X (43-58), Baby Boomer (59-77), and Silent Generation (78+).
- Environmental Concern, measured by a Likert-style item assessing participants' attitudes toward environmental issues when choosing products.

The entire survey was deployed through Qualtrics and distributed to a broad sample. The platform's randomizer function ensured each participant was shown only one version of the product description (either sustainability or durability framing). The questionnaire was structured to capture both the main framing effects and interactions with individual differences in values and demographics.

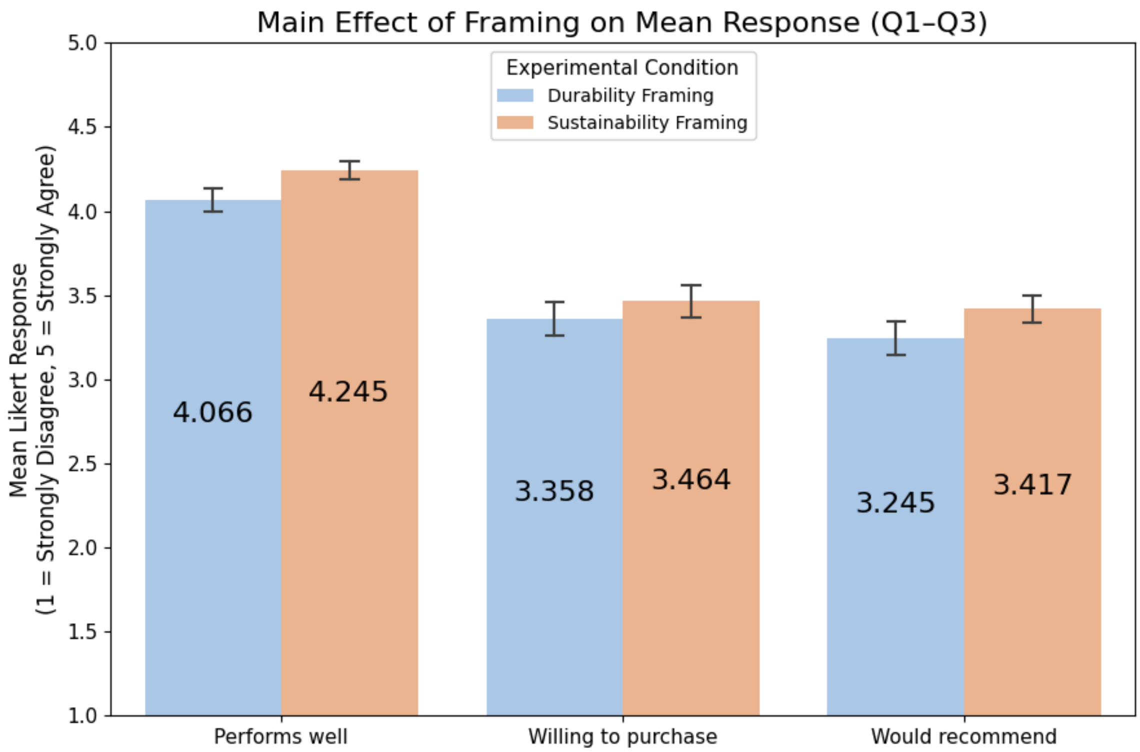
This design allowed us to isolate the causal effect of framing while also testing for nuanced moderation patterns. All responses were anonymized, and the data were exported for analysis in Python and R using regression and ANOVA techniques.

4. Results

4.1 T-test of Difference in Means

Our first analysis examined whether message framing influences consumer evaluations of the Stanley Quencher H2.0 Tumbler across three dimensions:

- **Q1**: *Perceived Performance*
- **Q2**: *Purchase Intention*
- **Q3**: *Recommendation Likelihood*



A two-sample t-test revealed that sustainability framing significantly improved perceived performance (Q1) compared to durability framing (M = 4.245 vs. M = 4.066; t = -2.03, p = 0.043). This suggests a "halo effect"; framing the product as eco-friendly enhances perceptions of quality, even if the product itself is unchanged.

However, there were no significant differences in purchase intention (Q2, p = 0.4539) or recommendation likelihood (Q3, p = 0.1805). This indicates that while sustainability messaging enhances perception, it does not necessarily drive action (purchase or advocacy), at least within this sample.

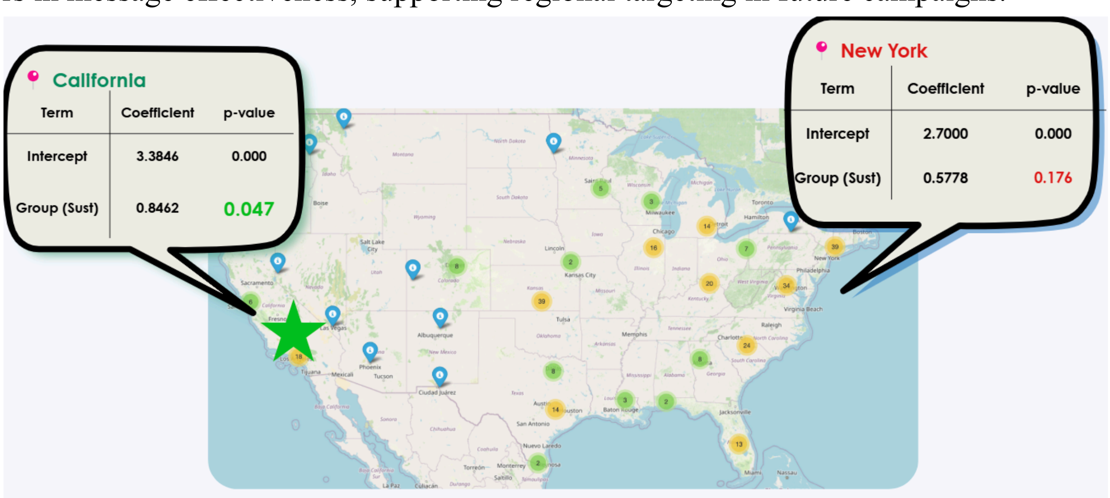
Durability vs. Sustainability Framing

Question	Perceived Performance	Purchase Intention	Recommendation Likelihood
n (Durability)	151	151	151
n (Sustainability)	151	151	151
Mean (Durability)	4.07	3.36	3.25
Mean (Sustainability)	4.25	3.46	3.42
t-stat	-2.033	-0.75	-1.342
p-value	**0.043**	0.4539	0.1805
Effect size d	-0.234	-0.086	-0.154
Power	0.526	0.116	0.268
Min Sample (for 0.8 power)	288	2109	659
Significant	**Yes**	No	No

Note: *The minimum sample size is calculated by G-Power*

Further, our power analysis suggests that much larger samples (e.g., 2,100 for purchase intention) would be required to detect smaller effects reliably. This reinforces the interpretation that while a trend exists, we may be underpowered to confirm it statistically.

Additionally, a regional analysis showed that California participants responded significantly more positively to sustainability messaging ($\beta = 0.85$, $p = 0.047$), while this framing had no significant effect in New York ($p = 0.176$). This suggests that geographic and cultural alignment matters in message effectiveness, supporting regional targeting in future campaigns.



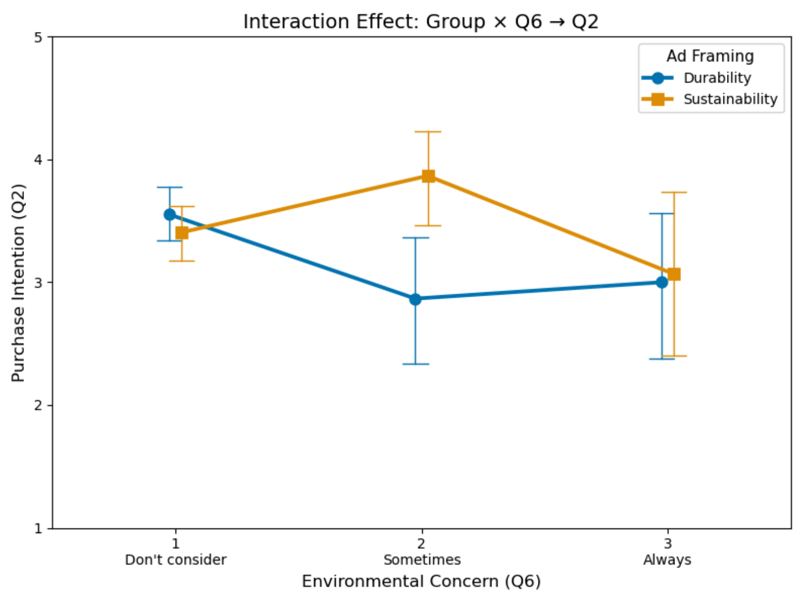
4.2 Two-way ANOVA

To assess whether environmental concern (Q6) moderates the effect of message framing on purchase intention (Q2), we conducted a two-way ANOVA. The interaction between Framing × Q6 was statistically significant ($F(2,109) = 5.27$, $p = 0.0056$), indicating that the effectiveness of sustainability messaging depends on environmental attitudes.

Framings × Environmental Concern(Q6) → Purchase Intention (Q2)

Source	Sum Sq	df	F	p-value
Framings	0.8011	1	0.55	0.459
Q6	5.4844	2	1.882	0.1542
Framings × Q6	15.3688	2	5.273	**0.0056**

The line chart shows that participants who “always” consider the environment (Q6 = 3) had significantly higher purchase intentions in the sustainability condition. In contrast, those who “never” or “sometimes” consider environmental issues showed little or opposite framing effects. This reveals a crossover interaction, sustainability framing works only when values align.



4.3 Three-way ANOVA

In our third analysis, we conducted a three-way ANOVA to investigate whether the effectiveness of ad framing depends not only on gender and age individually but also on their interaction. The results revealed a significant three-way interaction between Framing × Gender × Age Group (p = 0.0181), suggesting that the impact of sustainability messaging varies meaningfully across demographic subgroups.

Statistically, the interaction term Framings × Q4 (Age) × Q5 (Gender) had an F-value of 3.027 and a p-value of 0.018, indicating a reliable moderation effect. The interaction between age and gender alone was marginally significant (p = 0.0877), highlighting that framing adds further explanatory value to this demographic pairing.

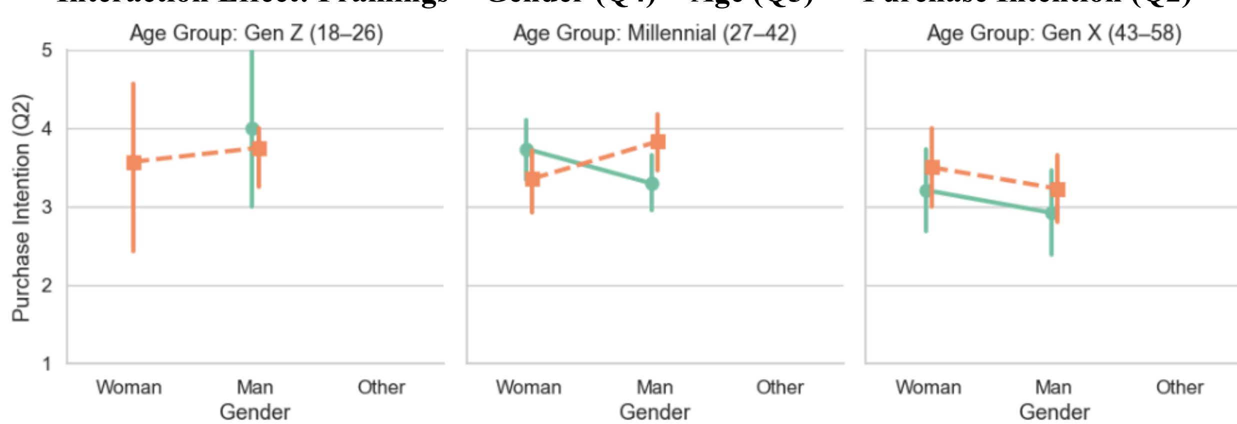
Framings × Gender (Q4) × Age (Q5) → Purchase Intention (Q2)

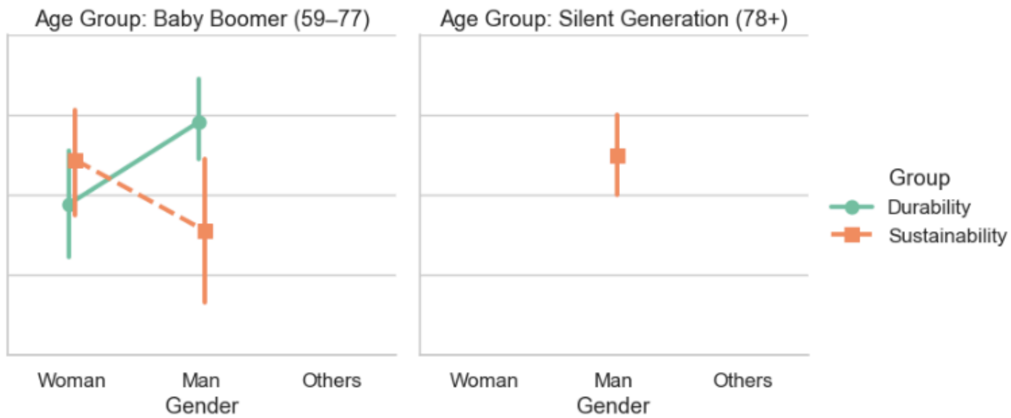
Source	Sum Sq	df	F	p-value
Intercept	74.068674	1.0	50.525255	0.0000001
Framings	0.306809	1.0	0.209287	0.6476752
Q4	5.768125	2.0	1.967336	0.1417232
Q5	7.125786	4.0	1.215197	0.3044714
Framings:Q4	1.567578	2.0	0.534654	0.4652574
Framings:Q5	4.985236	4.0	0.850158	0.4944370
Q4:Q5	22.753557	8.0	1.940141	0.08774697
Framings:Q4:Q5	35.503810	8.0	3.027324	**0.01810673**

As visualized in the plot, we excluded participants who selected “Other” for gender to focus on clearer group-level patterns. The plot illustrates purchase intention (Q2) by gender across five age groups, comparing the sustainability (orange dashed line) and durability (green solid line) frames. Notably, Millennial men and Baby Boomer women showed higher purchase intentions in the sustainability condition. Interestingly, both men and women in Gen X also showed increased purchase intent for the sustainability slogan.

These results suggest that sustainability messaging may resonate more strongly with specific demographic combinations. One potential explanation is that Millennial men may view sustainability as aligned with modern, progressive values, while Boomer women may associate it with caretaking and family well-being. This insight supports the value of tailoring marketing communications by audience segment to maximize relevance and impact.

Interaction Effect: Framings × Gender (Q4) × Age (Q5) → Purchase Intention (Q2)





5. Conclusion

Our study explored how different message framings, sustainability versus durability, shape consumer responses to the Stanley Cup. The results provide partial support for our hypotheses. Specifically, sustainability framing significantly enhanced perceived product performance, suggesting a halo effect where environmentally friendly messaging boosts impressions of quality. However, we did not find a clear effect of framing on purchase intention or recommendation likelihood, indicating that positive perception does not necessarily lead to action.

Importantly, our moderation analysis revealed that the effectiveness of sustainability framing depends on consumer characteristics, particularly environmental concerns. Participants with high environmental concern were more likely to express intent to purchase under sustainability framing, while those with lower concern showed no such effect. This finding emphasizes the importance of audience segmentation in marketing strategy. Geographic differences further support this point, California respondents were significantly more responsive to sustainability messaging than those in New York, likely reflecting regional value differences.

While our findings are insightful, the study has notable limitations. The sample size of approximately 300 participants constrained statistical power, especially when testing higher-order interactions. Power analyses suggest that much larger samples would be needed to detect smaller framing effects on behavioral outcomes like purchase intent. Additionally, our participant pool may lack the demographic diversity needed to generalize findings across broader populations.

In summary, while sustainability framing enhances perceptions of product quality, its influence on consumer behavior appears contingent on existing attitudes and demographics. Future research should address these limitations by using larger and more diverse samples and consider testing combined messaging strategies that highlight both ethical and functional benefits.

6. Appendix

6.1 AI Acknowledgement

Portions of this project, including data preprocessing, visualization, and grammar checking, were supported using AI assistance (ChatGPT by OpenAI). AI was used as a tool to enhance efficiency, ensure statistical clarity, and support the communication of results. All final interpretations and decisions were made by the research team.

6.2 R Code

The R code and outputs are in the separate Jupiter file.